PID Controller Design for Robust Performance

Ming-Tzu Ho and Chia-Yi Lin

***Abstract*—This note is devoted to the problem of synthesizing proportional–integral–derivative (PID) controllers for robust performance for a given single-input–single-output plant in the presence of uncertainty. First, the problem of robust performance design is converted into simultaneous stabilization of a complex polynomial family. An extension of the results on PID stabilization is then used to devise a linear programming design procedure for determining all admissible PID gain settings. The most important feature of the proposed approach is that it computationally characterizes the entire set of the admissible PID gain values for an arbitrary plant.**

***Index Terms*—Linear programming, proportional–integral–derivative (PID) control, robust performance.**

I. Introduction

The proportional–integral–derivative (PID) controller is the most widely used controller structure in industrial applications. Its structural simplicity and sufficient ability of solving many practical control problems have greatly contributed to this wide acceptance. Over the past decades, many PID design techniques [1] have been proposed for industrial use. Most of these design techniques are based on simple characterizations of process dynamics, such as the characterization by a first order model with time delay. In spite of this, for plants having higher order, there are very few generally accepted design methods existing. Robust performance design is one of fundamental problems in control. The problem of robust performance design is to synthesize a controller for which the closed-loop system is internally stabilized and the desired performance specifications are satisfied despite plant uncertainty. Although H_∞ [2] and μ-synthesis [3], [4] techniques have been successfully applied to solve the problem of robust performance design, design of the optimal or robust PID controller is a computationally intractable task [5] using H_∞ and μ-synthesis design techniques. Instead of directly using H_∞ or μ-synthesis design techniques, there were several approaches proposed to synthesize PID controllers for robust performance. For a given plant, [6] parametrized the stabilizing compensators that consist of a PID controller with a free parameter and the H_∞ design techniques were then used to select an appropriate free parameter to achieve robust performance. However, the order of the resulting PID-based compensator is always greater than the plant order. A parameter optimization approach was proposed in [7] and an LMI-based iterative optimization method was given in [8]. None of these guarantees global convergence with a reasonable amount of computation. By searching in the prescribed controller parameter space, [9] proposed a procedure for constructing the space of the admissible PID controller gain values for multiple performance specifications. Unfortunately, this method suffered from computational intractability. Based on gain and phase margin design, several simple tuning formulas for robust performance were given in [10] and [11]. These tuning formulas are limited to simple characterizations of process dynamics such as the characterization by a first-order model with time delay.

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The authors are with the Department of Engineering Science, National Cheng Kung University, Tainan 701, Taiwan (e-mail: bruceho@mail.ncku.edu.tw; n9690105@sparc1.cc.ncku.edu.tw).

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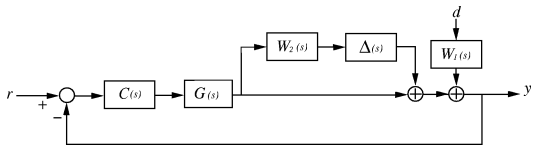


Fig. 1. Feedback control system with multiplicative uncertainty.

The aim of this note is to effectively solve the problem of robust performance PID design for an arbitrary plant. In particular, we focus on the problem of synthesizing a stabilizing PID controller, if any, for which the disturbance rejection design specification is achieved for a plant with multiplicative uncertainty. In the earlier work [12], based on the generalized Hermite–Biehler theorem [12], [13], a computational characterization of all stabilizing PID controllers was given for an arbitrary plant. This solution of the PID stabilization problem is an essential first step to any rational design of PID controllers. Recently, an extension of PID stabilization to the case of complex polynomials was developed in [14] and it was shown that such an extension could be exploited to carry out many H_∞ robust stability PID design problems. In this note, we show that the results from [14] can be also used to provide a computational characterization of all admissible PID controllers for robust performance. Such a characterization for all admissible PID controllers involves the solution of a linear programming problem. Accordingly, efficient algorithms are available for generating the parametric space of the entire admissible PID gain values. It will be clear from the exposition that PID design problems with other robust performance objectives can be treated in a similar fashion.

The note is organized as follows. In Section II, we show that the robust performance design problem of interest to us in this note can be converted into simultaneous stabilization of complex polynomials. The results of [14] can be then used for solving the resulting simultaneous stabilization problem. In Section III, we state the results of [14] on determining admissible real values of (k_p, k_i, k_d), if any, for which a complex polynomial of the form $L(s) + (k_d s^2 + k_p s + k_i)M(s)$ is Hurwitz, where $L(s)$ and $M(s)$ are some given complex polynomials. A linear programming characterization of all admissible (k_p, k_i, k_d) values is provided. These results immediately lead to a solution to the problem of synthesizing PID controllers for robust performance. In Section IV, a detailed synthesis procedure is presented and illustrated in a simple example. Finally, Section V contains some concluding remarks.

II. Robust Performance Design via Simultaneous Polynomial Stabilization

Consider the single-input–single-output feedback control system shown in Fig. 1. Here, r is the command signal, y is the output, and d is an energy-bounded disturbance. $G(s) = (N(s)/D(s))$ is the plant to be controlled, where $N(s)$ and $D(s)$ are coprime polynomials. $\Delta(s)$ is any stable and proper transfer function with $\|\Delta\|_\infty \leq 1$. The weights $W_1(s)$ and $W_2(s)$ describe the the frequency-domain characteristics of the performance specifications and model uncertainty, respectively. $C(s)$ is the controller used for making the closed-loop system stable and achieving desired design specifications. In this note, the controller $C(s)$ is chosen to be a PID controller, i.e.,

$$C(s) = k_p + \frac{k_i}{s} + k_d s = \frac{k_i + k_p s + k_d s^2}{s}. \tag{1}$$

Then, the complementary sensitivity function is

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

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and the sensitivity function is

$$S(s) = \frac{1}{1 + C(s)G(s)}.$$

Especially, we consider the problem of disturbance rejection for the plant with multiplicative uncertainty. This problem can be formulated as the following robust performance condition [15]:

$$\| |W_1(s)S(s)| + |W_2(s)T(s)| \|_\infty < 1. \tag{2}$$

Note that many other robust performance specifications can be formulated as (2). To convert the robust performance condition (2) into simultaneous polynomial stabilization, we first consider the following lemma.

Lemma 1: Let

$$\frac{A(s)}{B(s)} = \frac{a_0 + a_1 s + \cdots + a_x s^x}{b_0 + b_1 s + \cdots + b_x s^x}$$

and

$$\frac{E(s)}{F(s)} = \frac{e_0 + e_1 s + \cdots + e_y s^y}{f_0 + f_1 s + \cdots + f_y s^y}$$

be stable and proper rational functions with $b_x \neq 0$ and $f_y \neq 0$. Then

$$\left\| \left| \frac{A(s)}{B(s)} \right| + \left| \frac{E(s)}{F(s)} \right| \right\|_\infty < 1 \tag{3}$$

if and only if

a) $|a_x/b_x| + |e_y/f_y| < 1$;

b) $B(s)F(s) + e^{j\theta}A(s)F(s) + e^{j\phi}E(s)B(s)$ is Hurwitz for all θ and $\phi \in [0, 2\pi)$.

Proof: : Necessity Suppose that (3) holds. Then, condition a) is obvious. The necessity of condition b) is established in the following way. Since

$$\begin{aligned} &\left| \frac{A(j\omega)}{B(j\omega)} \right| + \left| \frac{E(j\omega)}{F(j\omega)} \right| \\ &\quad = \frac{|e^{j\theta}A(j\omega)F(j\omega)| + |e^{j\phi}E(j\omega)B(j\omega)|}{|B(j\omega)F(j\omega)|} \\ &\quad < 1 \end{aligned}$$

it follows that

$$|B(j\omega)F(j\omega)| > |e^{j\theta}A(j\omega)F(j\omega) + e^{j\phi}E(j\omega)B(j\omega)|.$$

Because $B(s)$ and $F(s)$ are Hurwitz, using Rouché's Theorem [16], we conclude that $B(s)F(s)+e^{j\theta}A(s)F(s)+e^{j\phi}E(s)B(s)$ is Hurwitz for all θ and $\phi \in [0, 2\pi)$.

Sufficiency Proceeding by contradiction, we assume that conditions a) and b) are true, but

$$\left\| \left| \frac{A(s)}{B(s)} \right| + \left| \frac{E(s)}{F(s)} \right| \right\|_\infty \geq 1.$$

Since $|A(j\omega)/B(j\omega)| + |E(j\omega)/F(j\omega)|$ is a continuous function of ω and

$$\lim_{\omega \to \infty} \left[\left| \frac{A(j\omega)}{B(j\omega)} \right| + \left| \frac{E(j\omega)}{F(j\omega)} \right| \right] = \left| \frac{a_x}{b_x} \right| + \left| \frac{e_y}{f_y} \right| < 1$$

then there must exist at least one $\omega_0 \in \mathcal{R}$ such that

$$\begin{aligned} &\left| \frac{A(j\omega_0)}{B(j\omega_0)} \right| + \left| \frac{E(j\omega_0)}{F(j\omega_0)} \right| \\ &= \frac{|A(j\omega_0)F(j\omega_0)| + |E(j\omega_0)B(j\omega_0)|}{|B(j\omega_0)F(j\omega_0)|} \\ &= 1. \end{aligned}$$

Therefore, it implies that there exist θ_1 and $\theta_2 \in [0, 2\pi)$ such that

$$B(j\omega_0)F(j\omega_0) + e^{j\theta_1}A(j\omega_0)F(j\omega_0) + e^{j\theta_2}E(j\omega_0)B(j\omega_0) = 0$$

and this obviously contradicts condition b). ■

Consider the stable weighting functions $W_1(s) = (N_{W1}(s)/D_{W1}(s))$ and $W_2(s) = (N_{W2}(s)/D_{W2}(s))$, where $N_{W1}(s)$, $D_{W1}(s)$, $N_{W2}(s)$, and $D_{W2}(s)$ are some real polynomials. Also, we denote the closed-loop characteristic polynomial to be

$$\rho(s, k_p, k_i, k_d) \triangleq sD(s) + (k_i + k_p s + k_d s^2)N(s).$$

For notational simplicity, we define the complex polynomial

$$\begin{aligned} &\psi(s, k_p, k_i, k_d, \theta, \phi) \\ &\quad \triangleq sD_{W1}(s)D_{W2}(s)D(s) \\ &\qquad + e^{j\theta}sN_{W1}(s)D_{W2}(s)D(s) \\ &\qquad + (k_d s^2 + k_p s + k_i)[D_{W1}(s)D_{W2}(s)N(s) \\ &\qquad + e^{j\phi}D_{W1}(s)N_{W2}(s)N(s)]. \end{aligned}$$

The objective of this note is to determine stabilizing PID controllers such that the robust performance condition (2) holds. Based on Lemma 1, the problem of synthesizing PID controllers for robust performance can be converted into the problem of determining values of (k_p, k_i, k_d) for which the following conditions hold:

1) $\rho(s, k_p, k_i, k_d)$ is Hurwitz;
2) $\psi(s, k_p, k_i, k_d, \theta, \phi)$ is Hurwitz for all θ and $\phi \in [0, 2\pi)$;
3) $|W_1(\infty)S(\infty)| + |W_2(\infty)T(\infty)| < 1$.

In view of these conditions, the problem of synthesizing PID controllers for robust performance has been expressed as simultaneous polynomial stabilization. With a fixed θ and a fixed ϕ, both $\psi(s, k_p, k_i, k_d, \theta, \phi)$ and $\rho(s, k_p, k_i, k_d)$ become in the form of $L(s) + (k_d s^2 + k_p s + k_i)M(s)$ where $L(s)$ and $M(s)$ are some given complex polynomials. The result of stabilizing the complex polynomial $L(s)+(k_d s^2+k_p s+k_i)M(s)$ is stated in Section III.

III. Characterization of All Admissible Gain Values for Extended PID Stabilization

Consider the complex polynomial of the form

$$\delta(s, k_p, k_i, k_d) = L(s) + (k_d s^2 + k_p s + k_i)M(s) \tag{4}$$

where $L(s)$ and $M(s)$ are arbitrary complex polynomials. In this section, we focus on the problem of determining those real values of (k_p, k_i, k_d), if any, for which (4) is Hurwitz stable. Note that by setting $L(s) = sD(s)$ and $M(s) = N(s)$, the problem stated above becomes PID stabilization. We, therefore, refer to stabilization of (4) as the extended PID stabilization problem. Based on a complex version of the generalized Hermite–Biehler theorem, [14] provided a computational characterization of all admissible (k_p, k_i, k_d) values for this extended PID stabilization problem. For detailed derivation, the reader is referred to [14].

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To set the stage for presenting this characterization, we first introduce some notion and definitions. The standard signum function $\mathrm{sgn} : \mathcal{R} \to \{-1, 0, 1\}$ is defined by

$$\mathrm{sgn}[x] = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0. \end{cases}$$

We consider a polynomial $\delta(s)$ of degree n

$$\delta(s) = \delta_0 + \delta_1 s + \delta_2 s^2 + \cdots + \delta_n s^n$$
$$\delta_i \in \mathcal{C}, \qquad i = 0, 1, \ldots, n, \quad \delta_n \neq 0.$$

Define the signature of the polynomial $\delta(s)$ denoted by $\sigma(\delta)$, as

$$\sigma(\delta) := \text{number of open left}-\text{half plane roots of } \delta(s) \\ -\text{number of open right}-\text{half plane roots of } \delta(s).$$

Let

$$L(s) = (a_0 + jb_0) + (a_1 + jb_1)s + \cdots + (a_{k-1} + jb_{k-1})s^{k-1} + (a_k + jb_k)s^k \qquad a_k + jb_k \neq 0$$
$$M(s) = (c_0 + jd_0) + (c_1 + jd_1)s + \cdots + (c_{m-1} + jd_{m-1})s^{m-1} + (c_m + jd_m)s^m \qquad c_m + jd_m \neq 0.$$

Consider the following "real-imaginary" decompositions of $L(s)$ and $M(s)$:

$$L(s) = L_R(s) + L_I(s)$$
$$M(s) = M_R(s) + M_I(s)$$

where

$$L_R(s) = a_0 + jb_1 s + a_2 s^2 + jb_3 s^3 + \cdots$$
$$L_I(s) = jb_0 + a_1 s + jb_2 s^2 + a_3 s^3 + \cdots$$
$$M_R(s) = c_0 + jd_1 s + c_2 s^2 + jd_3 s^3 + \cdots$$
$$M_I(s) = jd_0 + c_1 s + jd_2 s^2 + c_3 s^3 + \cdots.$$

Define

$$M^*(s) = M_R(s) - M_I(s).$$

Also, let n, m be the degrees of $\delta(s, k_p, k_i, k_d)$ and $M(s)$, respectively. Multiplying $\delta(s, k_p, k_i, k_d)$ by $M^*(s)$ and evaluating the resulting polynomial at $s = j\omega$, we obtain

$$\delta(j\omega, k_p, k_i, k_d)M^*(j\omega) = p(\omega, k_i, k_d) + jq(\omega, k_p)$$

where

$$p(\omega, k_i, k_d) = p_1(\omega) + (k_i - k_d\omega^2)p_2(\omega)$$
$$q(\omega, k_p) = q_1(\omega) + k_p q_2(\omega)$$
$$p_1(\omega) = L_R(j\omega)M_R(j\omega) - L_I(j\omega)M_I(j\omega)$$
$$p_2(\omega) = M_R^2(j\omega) - M_I^2(j\omega)$$
$$q_1(\omega) = \frac{1}{j}[L_I(j\omega)M_R(j\omega) - L_R(j\omega)M_I(j\omega)]$$
$$q_2(\omega) = \omega\left[M_R^2(j\omega) - M_I^2(j\omega)\right].$$

Also, define

$$q_f(\omega, k_p) = \frac{q(\omega, k_p)}{(1 + \omega^2)^{\frac{m+n}{2}}}.$$

Definition 1: Let $m, n, q_f(\omega, k_p)$ be as already defined. Denote ξ to be the leading coefficient of $\delta(s, k_p, k_i, k_d)M^*(s)$. For a given fixed k_p, let $\omega_1 < \omega_2 < \cdots < \omega_{l-1}$ be the real, distinct finite zeros of $q_f(\omega, k_p)$ with odd multiplicities. Also, define $\omega_0 = -\infty$ and $\omega_l = +\infty$. Define a sequence of numbers $i_0, i_1, i_2, \ldots, i_l$ as shown in the first equation at the bottom of the page, where the second equation shown at the bottom of the page holds. Next, we introduce the set $A_{k_p}(\gamma)$ of strings in A_{k_p} with a prescribed signature γ. To do so, we first need to define the signature $\gamma(\mathcal{I})$ associated with any element $\mathcal{I} \in A_{k_p}$. This definition is motivated by Theorem 1 to follow.

Definition 2: Let $m, n, \xi, q(\omega, k_p), q_f(\omega, k_p)$ be as already defined. For a given fixed k_p, let $\omega_1 < \omega_2 < \cdots < \omega_{l-1}$ be the real, distinct finite zeros of $q_f(\omega, k_p)$ with odd multiplicities. Also, define $\omega_0 = -\infty$ and $\omega_l = +\infty$. For each string $\mathcal{I} = \{i_0, i_1, \ldots, i_l\}$ or $\{i_1, i_2, \ldots, i_{l-1}\}$ in A_{k_p}, let $\gamma(\mathcal{I})$ denote the signature associated with the string $\mathcal{I}$ defined by the first equation shown at the bottom of the next page.

Definition 3: The set of strings in A_{k_p} with a prescribed signature $\gamma = \psi$ is denoted by $A_{k_p}(\psi)$. For a given fixed k_p, we also define the set of admissible strings for the extended PID stabilization problem as

$$F_{k_p}^* = A_{k_p}(n + \sigma(M^*(s))).$$

We are now ready to state the main result of this section.

Theorem 1: (Result on Extended PID Stabilization) [14] The extended PID stabilization problem, with a fixed k_p, is solvable for given complex polynomials $L(s)$ and $M(s)$ if and only if the following conditions hold.

i) $F_{k_p}^*$ is not empty where $F_{k_p}^*$ is as already defined, i.e., at least one feasible string exists and

$$A_{k_p} := \begin{cases} \{\{i_0, i_1, \ldots, i_l\}\}, & \text{if } m + n \text{ is even and } \xi \text{ is purely real, or} \\ & m + n \text{ is odd and } \xi \text{ is purely imaginary.} \\ \{\{i_1, i_2, \ldots, i_{l-1}\}\}, & \text{if } m + n \text{ is even and } \xi \text{ is not purely real, or} \\ & m + n \text{ is odd and } \xi \text{ is not purely imaginary.} \end{cases}$$

$$\text{for } t = 0, 1, \ldots, l, \qquad i_t \in \begin{cases} \{-1, 1\}, & \text{if } j\omega_t \text{ is not a } j\omega-\text{axis root of } M^*(s) \\ \{0\}, & \text{otherwise.} \end{cases}$$

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ii) There exists a string $\mathcal{I}$, either $\mathcal{I} = \{i_0, i_1, \ldots, i_l\}$ or $\{i_1, i_2, \ldots, i_{l-1}\} \in F^*_{k_p}$, and values of k_i and k_d such that either $\forall t = 0, 1, \ldots, l$ or $1, 2, \ldots, l-1$ for which $M^*(j\omega_t) \neq 0$

$$\left[p_1(\omega_t) + \left(k_i - k_d\omega_t^2\right) p_2(\omega_t)\right] i_t > 0 \tag{5}$$

where $p(\omega, k_i, k_d)$ is as already defined. Furthermore, if there exist values of k_i and k_d such that the above condition is satisfied for the feasible strings $\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_s \in F^*_{k_p}$, then the set of stabilizing (k_i, k_d) values corresponding to the fixed k_p is the union of the (k_i, k_d) values satisfying (5) for $\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_s$.

From Theorem 1, for a fixed k_p, the characterization of all stabilizing (k_i, k_d) values involves the feasible solution of the set of linear inequalities (5). Thus, linear programming can be used to characterize the region of all stabilizing (k_i, k_d) values. The set of all stabilizing (k_p, k_i, k_d) values can now be found by simply sweeping over k_p and determining the stabilizing (k_i, k_d) values by using Theorem 1 at each stage. The necessary condition for existence of the feasible string $\mathcal{I}$ such that $\gamma(\mathcal{I}) = n + \sigma(M^*(s))$ can be determined by the number of real and distinct finite zeros of $q_f(\omega, k_p)$. Hence the necessary ranges of k_p values over which the sweeping has to be carried out can then be *a priori* narrowed down by determining the ranges of k_p such that $q_f(\omega, k_p)$ has at least (see the second equation at the bottom of the page) real, distinct finite zeros with odd multiplicities. The resulting ranges of k_p are the only ranges of k_p for which stabilizing (k_i, k_d) values may exist. Using root locus ideas, [12] proposed a procedure for determining the real root distribution of $q(\omega, k_p) = q_1(\omega) + k_p q_2(\omega)$ corresponding to the different ranges of k_p.

IV. Robust Performance Design of PID Controllers

In this section, we show how the results of the last section can be used for carrying out robust performance PID design. As shown in Section II, we have that the problem of robust performance PID design is solvable if and only if there exists a gain value (k_p, k_i, k_d) such that condition (1), (2) and (3) hold. For condition (1), with a fixed k_p, we can set $L(s) = sD(s), M(s) = N(s)$ and then use the results from Theorem 1 for determining the stabilizing (k_i, k_d) gain values. The set of the stabilizing (k_i, k_d) gain values is denoted by $\mathcal{S}_{(1,k_p)}$. With the same k_p and any fixed θ and $\phi \in [0, 2\pi)$, we set $L(s) = sD_{W1}(s)D_{W2}(s)D(s) + e^{j\theta}sN_{W1}(s)D_{W2}(s)D(s)$ and $M(s) = D_{W1}(s)D_{W2}(s)N(s) + e^{j\phi}D_{W1}(s)N_{W2}(s)N(s)$, then again use the results from Theorem 1 for determining the set of admissible (k_i, k_d) gain values. This admissible set is denoted by $\mathcal{S}_{(2,k_p,\theta,\phi)}$. We now denote the set of admissible (k_i, k_d) gain values satisfying condition (2) to be $\mathcal{S}_{(2,k_p)}$. Then $\mathcal{S}_{(2,k_p)}$ is given by

$$\mathcal{S}_{(2,k_p)} = \cap_{\theta,\phi\in[0,2\pi)}\mathcal{S}_{(2,k_p,\theta,\phi)}.$$

Again with the same fixed k_p, the admissible (k_i, k_d) gain values satisfying condition (3) can be obtained by solving a linear programming problem and this admissible set is denoted by $\mathcal{S}_{(3,k_p)}$. Thus, for a fixed k_p, the set of all admissible (k_i, k_d) values for which the robust performance condition (2) is satisfied, denoted by $\mathcal{S}_{(k_p)}$ is given by

$$\mathcal{S}_{(k_p)} = \cap_{i=1,2,3}\mathcal{S}_{(i,k_p)}.$$

The set of all admissible (k_p, k_i, k_d) values such that the robust performance condition (2) is satisfied can now be found by simply sweeping over he necessary range of k_p and determining $\mathcal{S}_{(k_p)}$ at each stage.

To characterize the entire admissible set, the proposed design procedure requires to solve the complex PID stabilization problem by sweeping over the necessary ranges of $k_p, \theta \in [0, 2\pi)$ and $\phi \in [0, 2\pi)$. Instead of exhaustive search, k_p, θ and ϕ have to be discretized. Consequently, due to discretization of θ and ϕ, the resulting admissible set $\mathcal{S}_{(2,k_p)}$ is overbounded. Discretization of k_p could introduce inherent conservatism into the admissible (k_p, k_i, k_d) gain space. Although the method given in [9] can also be used to construct the entire set of admissible PID gain values, the approach proposed in this note is more efficient due to superior computational efficiency of its linear programming characterization.

Remark 1: It can be easily shown that for a fixed k_p the set of admissible (k_i, k_d) gain values such that $\||W_1(s)S(s))| + |W_2(s)T(s))|\|_\infty < 1$ is a union of convex sets. It has been shown that some robust controller design techniques could suffer from the fragility problem [17]. With this convexity of the admissible (k_i, k_d) gain values, the design procedure proposed in [18] can be used to carry out a nonfragile PID design in the controller parameter space.

Remark 2: The design procedure presented here is only applicable to the plants described by rational transfer functions. For the plant with time delay, the time delay factor can be approximated by a properly chosen Padé approximation. The approximation error can be treated as model uncertainty. Based on the resulting rational transfer function model, the proposed design procedure can then be carried out directly.

$$\gamma(\mathcal{I}) := \begin{cases} \frac{1}{2}\{i_0 \cdot (-1)^{l-1} + 2\sum_{r=1}^{l-1} i_r \cdot (-1)^{l-1-r} - i_l\} \cdot \operatorname{sgn}[q(\infty, k_p)], \\ \text{if } m+n \text{ is even and } \xi \text{ is purely real, or } m+n \text{ is odd and } \xi \text{ is purely imaginary.} \\ \frac{1}{2}\{2\sum_{r=1}^{l-1} i_r \cdot (-1)^{l-1-r}\} \cdot \operatorname{sgn}[q(\infty, k_p)], \\ \text{if } m+n \text{ is even and } \xi \text{ is not purely real, or } m+n \text{ is odd and } \xi \text{ is not purely imaginary.} \end{cases}$$

$$\begin{cases} |n + \sigma(M^*(s))| - 1, & \text{if } m+n \text{ is even and } \xi \text{ is purely real, or } m+n \text{ is odd and } \xi \text{ is purely imaginary} \\ |n + \sigma(M^*(s))|, & \text{if } m+n \text{ is even and } \xi \text{ is not purely real, or } m+n \text{ is odd and } \xi \text{ is not purely imaginary} \end{cases}$$

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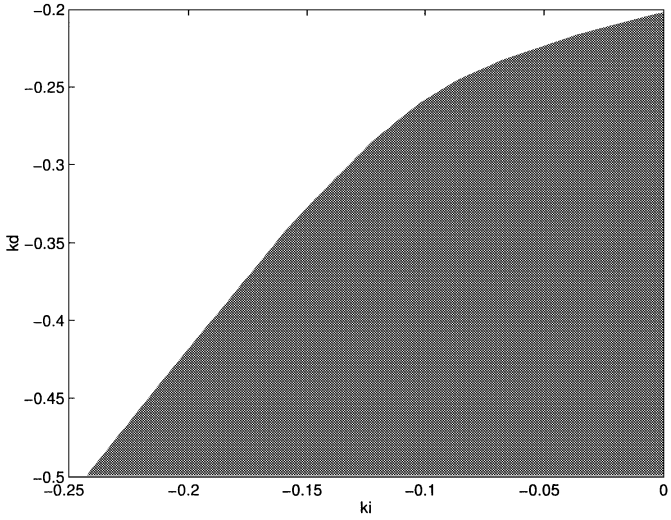


Fig. 2. Admissible set $\mathcal{S}_{(-0.3)}$.

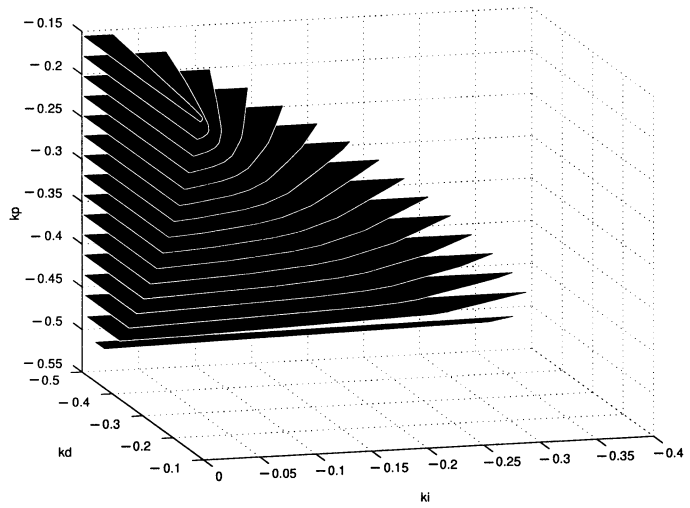


Fig. 3. Admissible set of (k_p, k_i, k_d) values for which $\||W_1(s)S(s,k_p,k_i,k_d)| + |W_2(s)T(s,k_p,k_i,k_d)|\|_\infty < 1$.

It is reasonable to expect that the higher the order of the Padé approximation chosen, the less conservative the result obtained.

Remark 3: To prevent noise amplification and internal instability, in practical applications, PID controllers are usually implemented in the following form:

$$C(s) = k_p + \frac{k_i}{s} + \frac{k_d s}{1+Ts} \tag{6}$$

where T is a small positive number. Using a linear transformation proposed in [12] to reparametrize the controller parameters of (6), the proposed design procedure is also applicable to the case of the modified PID structure (6).

Example 1: Consider the plant $G(s) = (s-15)/(s^2+s-1)$ and the PID controller $C(s) = (k_d s^2 + k_p s + k_i)/(s)$. Then, the sensitivity function and complementary sensitivity function are

$$\begin{aligned} &S(s,k_p,k_i,k_d) \\ &\quad = \frac{s(s^2+s-1)}{s(s^2+s-1)+(k_d s^2+k_p s+k_i)(s-15)} \\ &T(s,k_p,k_i,k_d) \\ &\quad = \frac{(k_d s^2+k_p s+k_i)(s-15)}{s(s^2+s-1)+(k_d s^2+k_p s+k_i)(s-15)}. \end{aligned}$$

The weighting functions are chosen as: $W_1(s) = (0.2)/(s+0.2)$ and $W_2(s) = (s+0.1)/(s+1)$. From Section II, we know that the admissible (k_p, k_i, k_d) values exist if and only if the following conditions hold:

1) $\rho(s,k_p,k_i,k_d) = s(s^2+s-1)+(k_d s^2+k_p s+k_i)(s-15)$ is Hurwitz;
2) $\psi(s,k_p,k_i,k_d,\theta,\phi) = s(s+0.2)(s+1)(s^2+s-1) + e^{j\theta}s(0.2)(s+1)(s^2+s-1)+(k_d s^2+k_p s+k_i)[(s+0.2)(s+1)(s-15)+e^{j\phi}(s+0.2)(s+0.1)(s-15)]$ is Hurwitz for all θ and $\phi \in [0, 2\pi)$;
3) $|W_1(\infty)S(\infty,k_p,k_i,k_d)| + |W_2(\infty)T(\infty,k_p,k_i,k_d)| = |k_d/(k_d+1)| < 1$.

With a fixed k_p, for instance $k_p = -0.3$, and following the design procedure stated before, we obtain the admissible set $\mathcal{S}_{(-0.3)}$ sketched in Fig. 2. It verifies the convexity of admissible (k_i, k_d) values for a fixed k_p. Now, using root locus ideas [12], it was determined that a necessary condition for the existence of admissible (k_i, k_d) values is that $k_p \in (-0.5079, -0.1155)$. Thus, by sweeping over $k_p \in (-0.5079, -0.1155)$, and using the aforementioned design procedure, we obtained the admissible set of (k_p, k_i, k_d) values for which $\||W_1(s)S(s,k_p,k_i,k_d)| + |W_2(s)T(s,k_p,k_i,k_d)|\|_\infty < 1$. The entire admissible set is sketched in Fig. 3.

V. Concluding Remarks

In this note, we have shown how robust performance design can be converted into simultaneous stabilization of complex polynomials. By employing the results from the earlier work, a linear programming characterization of all admissible robust performance PID controllers has been provided. The most important feature of the proposed approach is that it computationally characterizes the entire set of the admissible PID gain values for an arbitrary plant. This characterization can facilitate the optimal design of any additional design specifications. The results presented in this note focused on a particular design specification but one can easily see that other robust performance PID design problems can be formulated in a similar manner and solved.

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Solution Bounds of the Continuous Riccati Matrix Equation

Chien-Hua Lee

***Abstract*—A new approach is proposed for estimating the solution of the continuous algebraic Riccati equation (CARE). Upper and lower solution bounds of the CARE are presented. Comparisons show that the present bounds are more general and/or tighter than existing results.**

***Index Terms*—Continuous algebraic Riccati equation, eigenvalue bounds, matrix bounds.**

I. Introduction

Consider the continuous algebraic Riccati equation (CARE)

$$PA + A^T P - PBB^T P = -Q \tag{1}$$

where $A \in \Re^{n\times n}$ represents a constant matrix, $(^T)$ denotes the transpose, $Q \in \Re^{n\times n}$ is a positive–definite symmetric matrix, the matrix $P \in \Re^{n\times n}$ is the unique positive–definite symmetric solution of the CARE (1). It is assumed that (A, B) is a controllable pair and $(A, Q^{1/2})$ is an observable pair.

The Riccati and Lyapunov equations are usually utilized to solve the analysis and design problems for control systems. Sometimes, in order to reduce computational burdens, we need only bounds but not the exact solutions. Furthermore, the solution bounds of these equations can also be applied to deal with many control problems such as stability analysis for perturbed systems subjected to time delay(s) [19], [23], robust root clustering analysis [18], [25], determination of the size of the estimation error for multiplicative systems [9], as well as analytical and synthetic problems involving these equations [21]. Therefore, during the past two decades, a number of works have been proposed for dealing with the estimation problem for the solutions of these equations. They estimated bounds of the extreme eigenvalues, the trace, the determinant, and the matrix for the solutions [1]–[8], [11]–[17], [22], [24]. Of those estimations, all eigenvalue bounds can be obtained directly from the matrix bounds. Hence, the latter are the most general findings. However, most of researches discussed eigenvalue bounds. Few works focus on the estimation problem for the solution matrices. Surveying the literature, it seems that there are only [1], [16], and [17] have developed the matrix bounds for the solution of the CARE. Therefore, this note discusses upper and lower matrix bounds of the solution of this equation. By a new approach, several matrix bounds are proposed. Comparisons show that existing results reported in [1], [16], and [17] are only special cases of this work. Therefore, this work can be considered as a generalization for previous results. Furthermore, the proposed bounds are also sharper than those presented in [16] and [17].

The following symbol conventions are used in this note. $\Re$ denotes the real number field. The inequality $A > (\geq) B$ means $A-B$ is a positive (semi)definite symmetric matrix; $\lambda_i(A)$ denotes the ith eigenvalue of a symmetric matrix A for $i = 1, 2, \ldots, n$ whereas $\lambda_i(A)$ is arranged in the nonincreasing order (i.e., $\lambda_1(A) \geq \lambda_2(A) \geq \cdots \geq \lambda_n(A)$). The identity matrix with appropriate dimensions is represented by I.

II. Main Results

Before developing the main results, we first review the following useful result.

Lemma 1 [10]: For any symmetric matrices A and B, the following inequality:

$$\begin{bmatrix} A & D \\ D^T & B \end{bmatrix} > 0 \tag{2}$$

is equivalent to either

$$A > 0 \quad B - D^T A^{-1} D > 0 \tag{3}$$

or

$$B > 0 \quad A - D^T B^{-1} D > 0 \tag{4}$$

where D is any constant matrix with appropriate dimension.

Utilizing the aforementioned useful result, the main results of this note are derived as follows.

Theorem 1: Let P be the positive solution of the CARE (1). Then P has the lower bound

$$P \geq G_1^{-1}(G_1(Q - A^T R_1 A)G_1)^{1/2} G_1^{-1} \equiv P_{l1} \tag{5}$$

where the positive–definite matrix G_1 is defined by

$$G_1 \equiv \left(BB^T + R_1^{-1}\right)^{1/2} \tag{6}$$

and the positive constant matrix R_1 is chosen as $Q > A^T R_1 A$.

Furthermore, if the matrix BB^T is nonsingular, then P has the upper matrix bound

$$P \leq E_1^{-1}(E_1(Q + A^T \bar{R}_1 A)E_1)^{1/2} E_1^{-1} \equiv P_{u1} \tag{7}$$

where the positive–definite matrix $\bar{R}_1$ is selected such that

$$E_1 \equiv \left(BB^T - \bar{R}_1^{-1}\right)^{1/2} > 0. \tag{8}$$

Proof: Define a positive–semidefinite matrix Φ as

$$\begin{aligned} \Phi &\equiv \left(R_1^{-1/2} P + R_1^{1/2} A\right)^T \left(R_1^{-1/2} P + R_1^{1/2} A\right) \\ &= PR_1^{-1} P + PA + A^T P + A^T R_1 A \geq 0 \end{aligned} \tag{9}$$

where R_1 is a positive constant matrix. Using (9), the CARE (1) becomes

$$\begin{aligned} A^T P + PA + Q &= \Phi - PR_1^{-1} P - A^T R_1 A + Q \\ &= PBB^T P. \end{aligned} \tag{10}$$

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The author is with the Department of Electrical Engineering, Cheng-Shiu University, Taiwan 833, R.O.C.

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